MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, MODULAR, TYPE IV, CONNECTOR, 0.3 PITCH, 100 CONTACT, OFFSET

This specification is approved for use within Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of the following specification listed in that issue of the Department of Defense Index of Specifications (DODISS) specified in the solicitation: MIL-C-28754.

AMSC N/A FSC 5935

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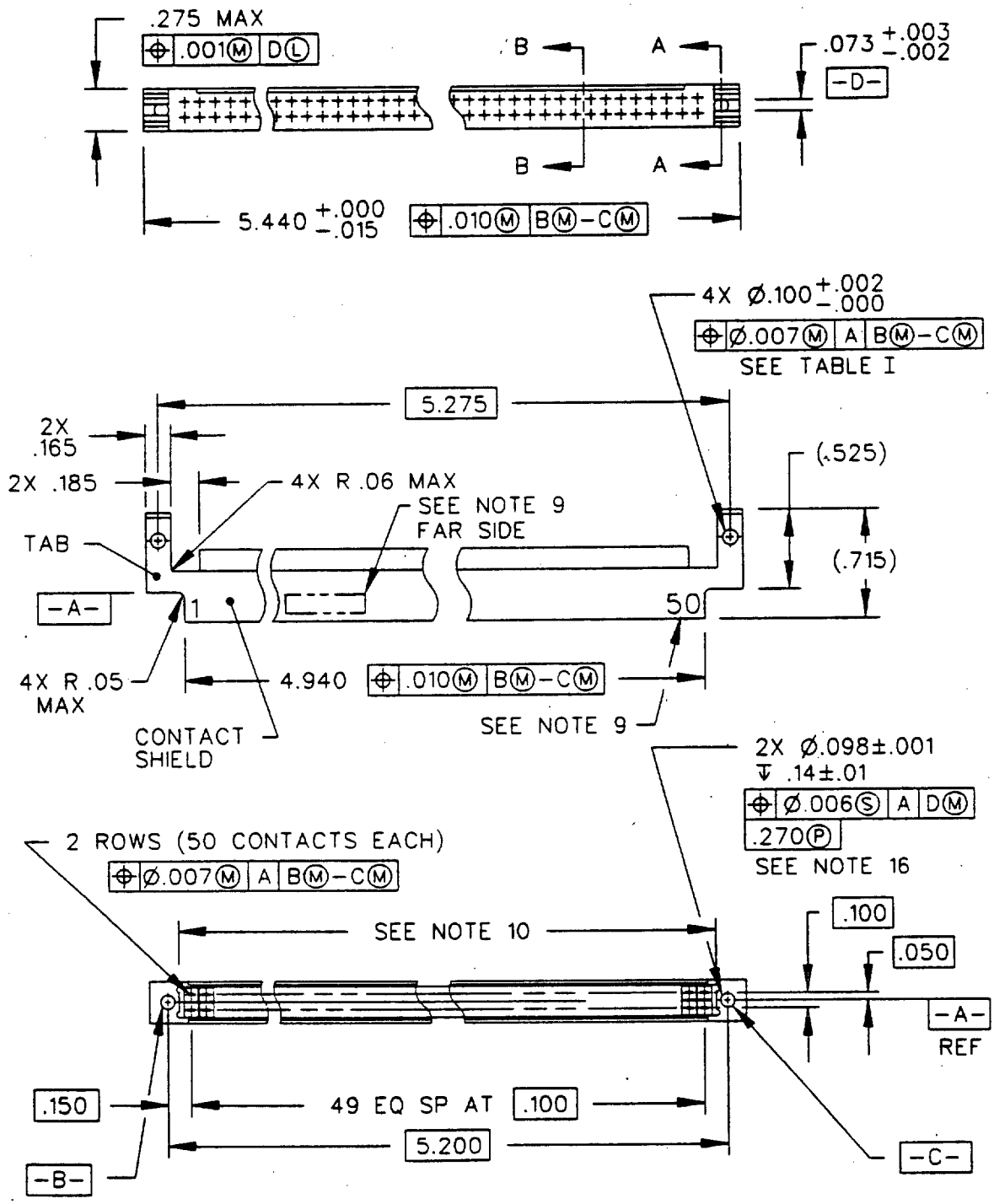
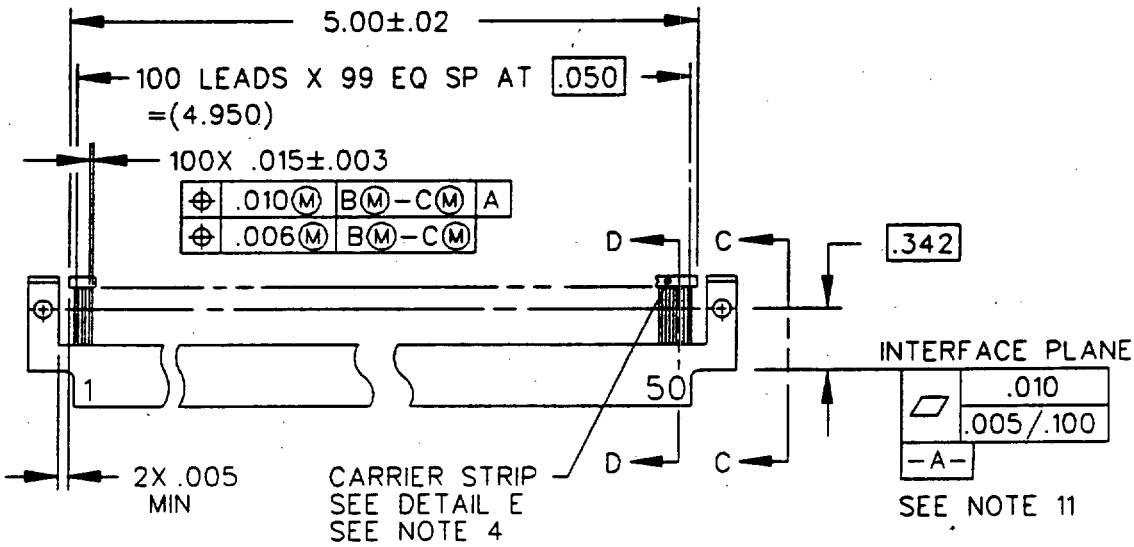
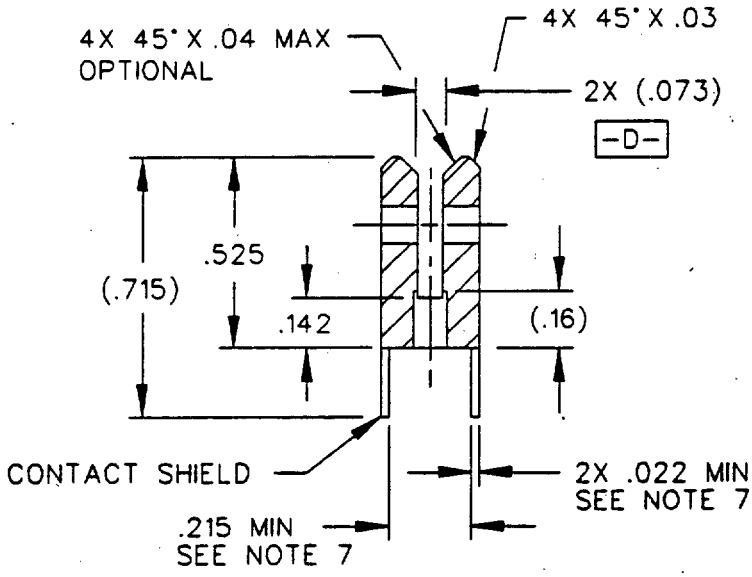


FIGURE 1. Dimensions and configuration.

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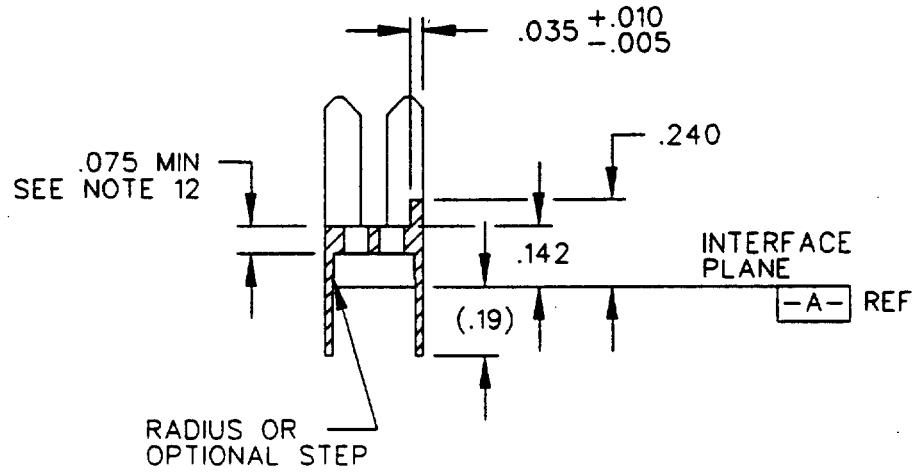




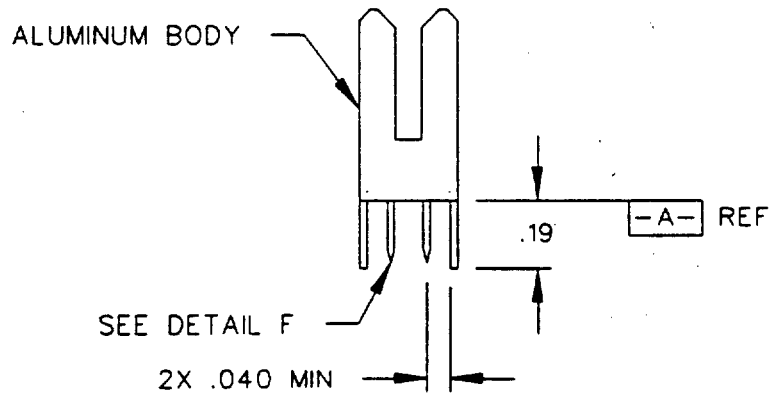
SECTION A-A
ROTATED 90° CW

FIGURE 1. Dimensions and configuration - Continued.

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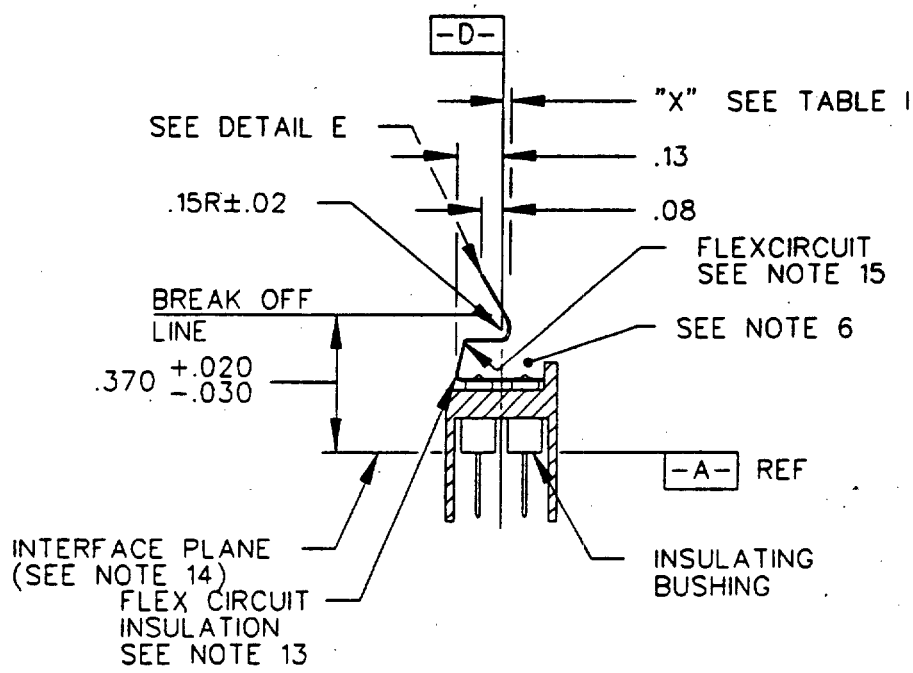
SECTION B−B
ROTATED 90° CW



VIEW C−C

FIGURE 1. Dimensions and configuration - Continued.

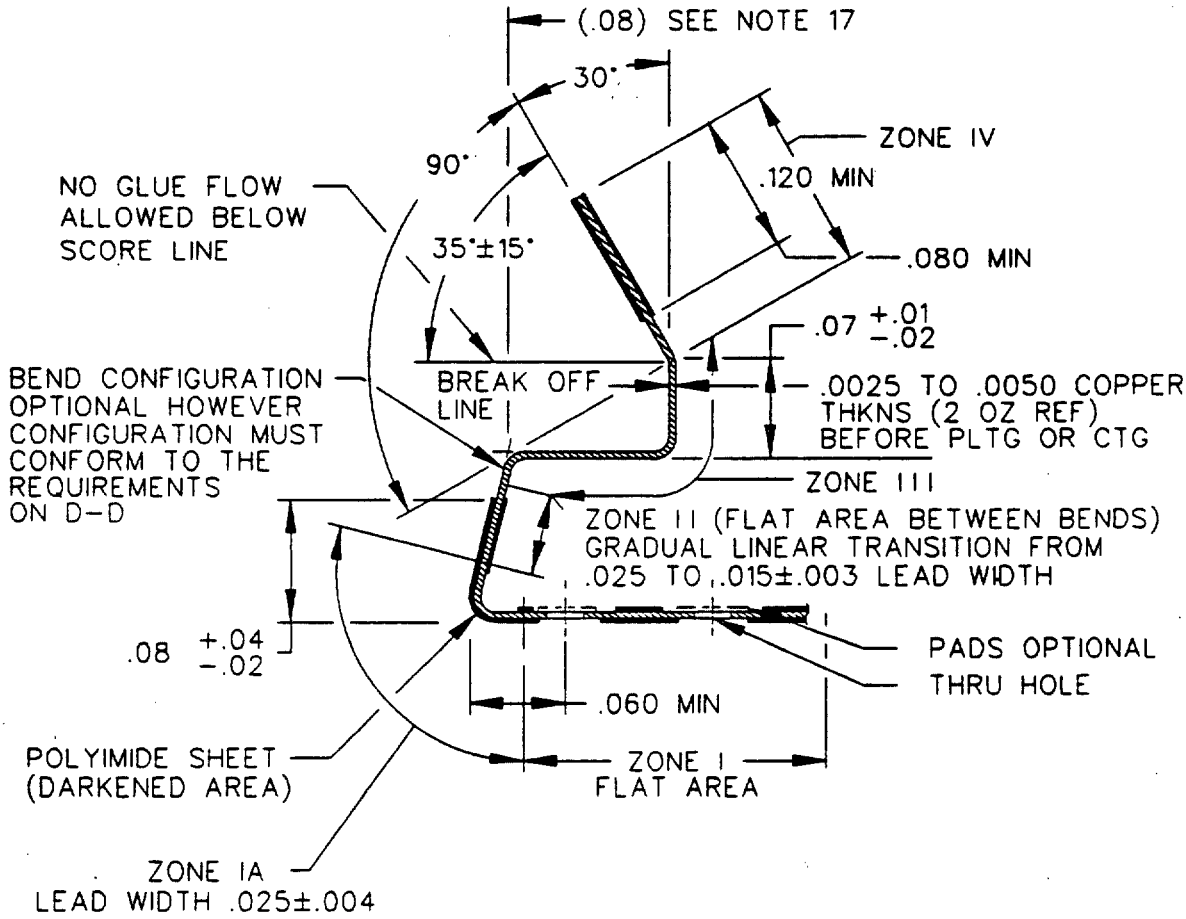
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SECTION D-D

FIGURE 1. Dimensions and configuration - Continued.

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FLEX CIRCUIT CROSS-SECTION
(AFTER FORMING)

DETAIL E

FIGURE 1. Dimensions and configuration - Continued.

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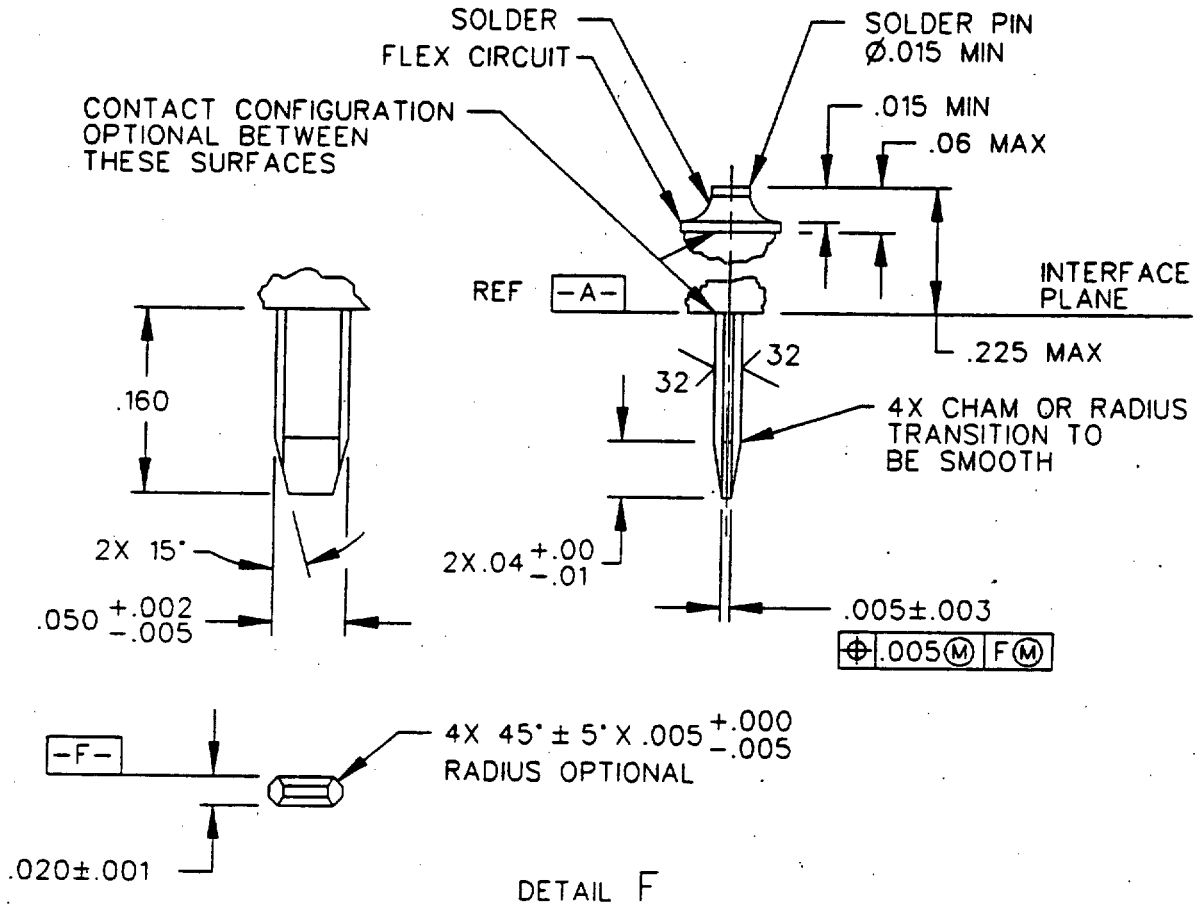
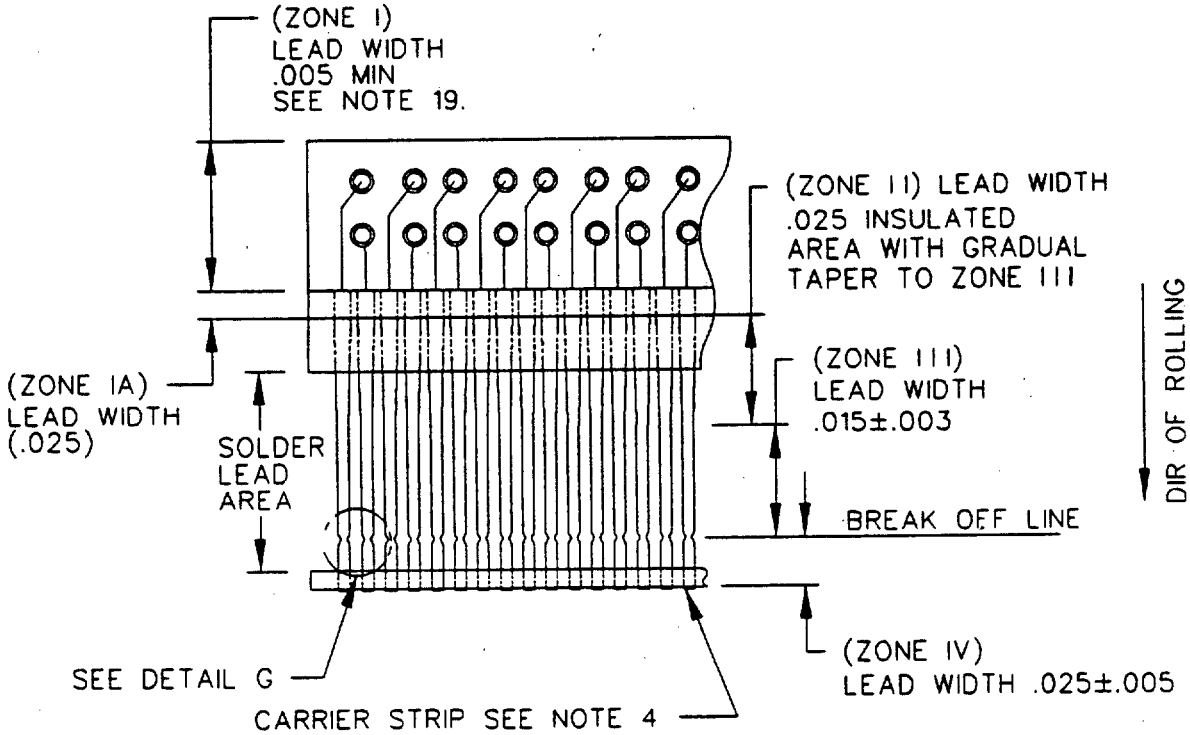


FIGURE 1. Dimensions and configuration - Continued.

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FLEX CIRCUIT CONFIGURATION
(BEFORE FORMING)

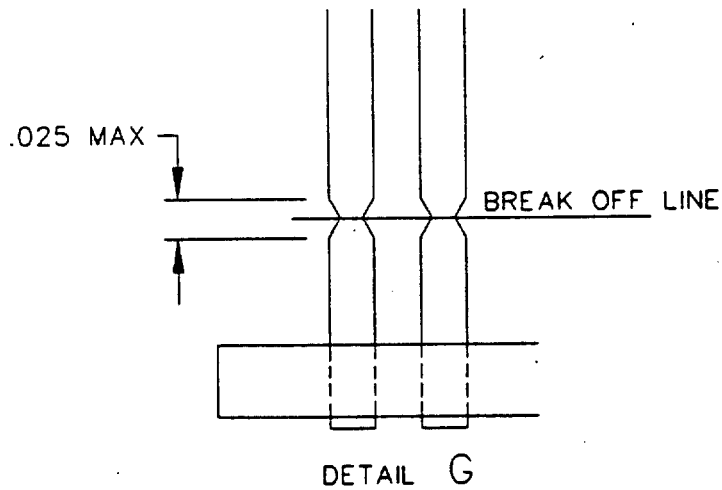
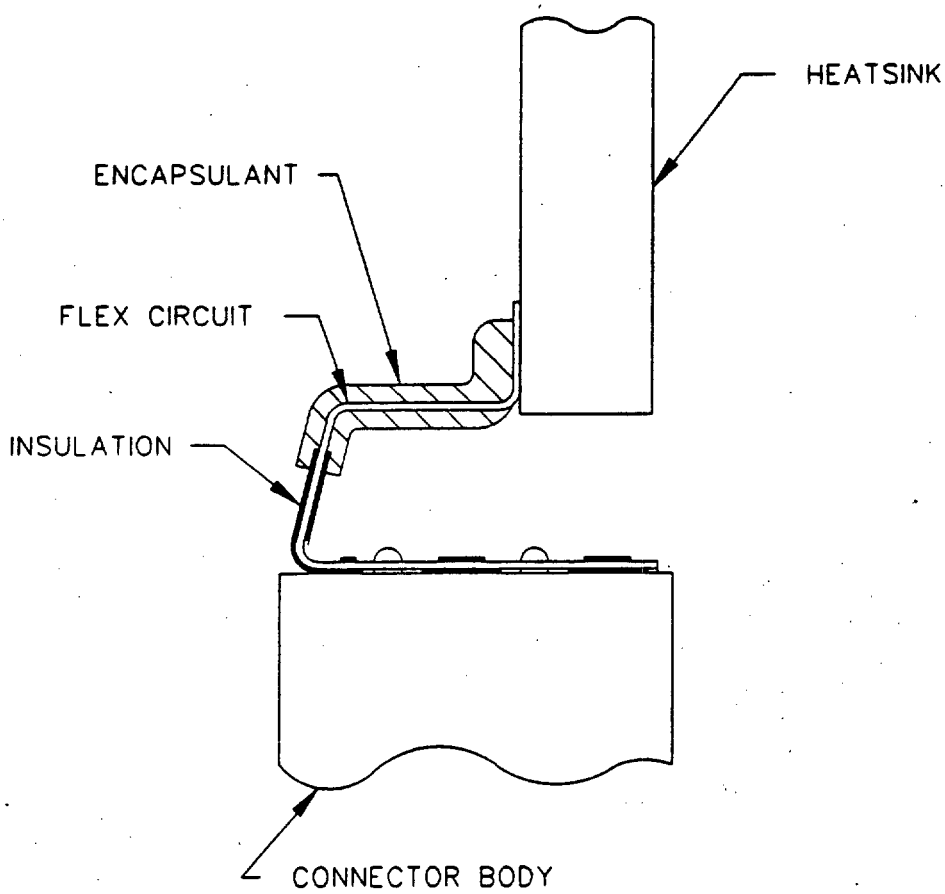


FIGURE 1. Dimensions and configuration - Continued.

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FLEX CIRCUIT CROSS-SECTION
(ASSEMBLED MODULE)

FIGURE 1. Dimensions and configuration - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
0.000032	0.0008	0.022	0.56	0.100	2.54	0.270	6.86
0.001	0.03	0.025	0.64	0.130	3.30	0.275	6.99
0.002	0.05	0.030	0.76	0.142	3.60	0.342	8.69
0.0025	0.0635	0.035	0.89	0.150	3.81	0.370	9.40
0.003	0.08	0.040	1.02	0.160	4.06	0.505	12.83
0.005	0.13	0.050	1.27	0.165	4.19	0.525	13.34
0.006	0.15	0.060	1.52	0.167	4.24	0.715	18.16
0.007	0.18	0.063	1.60	0.185	4.70	4.940	125.48
0.008	0.20	0.070	1.78	0.190	4.83	4.950	125.73
0.010	0.25	0.073	1.86	0.215	5.46	5.000	127.00
0.015	0.38	0.075	1.91	0.225	5.72	5.200	132.08
0.018	0.46	0.080	2.03	0.230	5.84	5.275	133.99
0.020	0.51	0.098	2.49	0.240	6.10	5.440	138.18

NOTES:

1. Dimensions are in inches.

2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

3. Unless otherwise specified, tolerances are plus or minus 0.005 (0.13 mm) inch for three place decimals, plus or minus 0.01 (0.3 mm) inch for two place decimals and plus or minus 5° for angles.

4. The carrier strip shall remain attached to the contact tails to facilitate shipping. It shall be scored or reduced at the point indicated in detail F to facilitate removal at final assembly. The carrier strip shall be a non-conductive material. The width of the strip shall be such that there is no adhesive flow into the 0.07 lead area.

FIGURE 1. Dimensions and configuration - Continued.

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5. Dimensional limits apply after plating and anodizing.

6. Clearance between top of solder pins and bottom of solder lead shall be 0.030 (0.76 mm) minimum. The projection of the solder pins through the flexcircuit shall be 0.015 (0.38 mm) minimum.

7. Dimension applies below interface plane.

8. Solder lead spacing and locational tolerance applies only for the 0.07 (1.78 mm) dimension below the score line. See Detail E.

9. Mark contact numbers with 0.12 (3.0 mm) plus or minus 0.02 (0.5 mm) inch high characters using white epoxy ink per MIL-I-43553. Mark corresponding contact numbers on opposite contact shield. Mark part number, manufacturer's code ident and date code in this area with 0.080 (2.0 mm) plus or minus 0.020 (0.5 mm) inch high characters using white epoxy ink per MIL-I-43553.

10. Length of recessed area shall be adequate to allow proper installation of insulator bushings and shall maintain a 0.030 (0.76 mm) minimum radial wall thickness around the 0.098 (2.5 mm) diameter holes. Form of recess adjacent to holes is optional.

11. Interface plane consists of the individual insulator bushing and the raised surface of the connector body adjacent to the 0.098 holes. Depth of the recessed area shall be controlled such that all bushings must be fully seated and meet the flatness requirement specified for the interface plane. The bushing for the grounding feature is exempt from the interface plane flatness requirement and may be recessed below the interface plane. It shall not protrude above the interface plane.

12. Body material thickness.

FIGURE 1. Dimensions and configuration - Continued.

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13. Flexcircuit insulation is required on the bend area. Flexcircuit shall not extend beyond connector profile. Insulation on inside of flexcircuit is optional.

14. Plating defects on the carrier strip above the break-off line are acceptable, provided that inspection testing of the connector is not effected.

15. Flexcircuit shall have a minimum bend radii of 0.010 inch.

16. The bottom 0.01 inch of the hole depth can have a reduced diameter of 0.02 inch maximum.

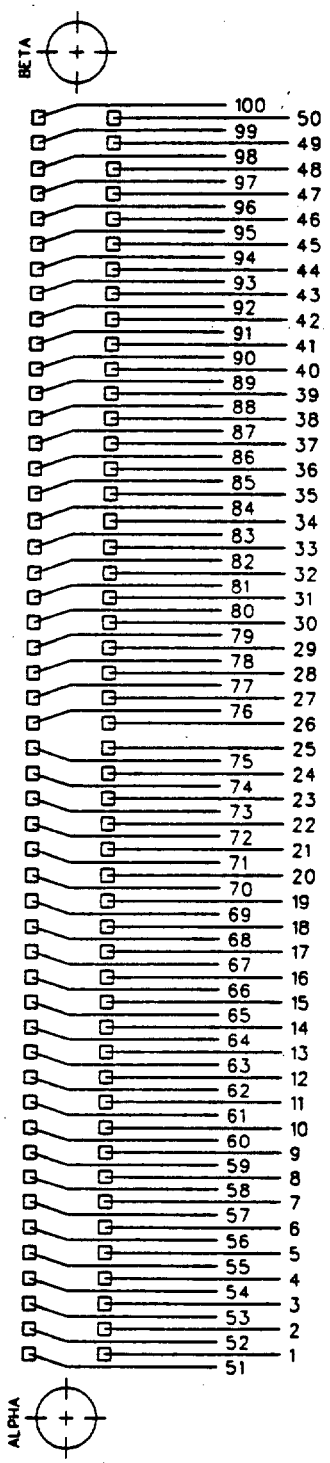
17. The 0.08 dimension shall be measured from point of intersection of the tangent lines of Zone II and the lower portion of Zone III as shown in flex circuit cross-section (after forming).

18. The assembled connector shall conform to all applicable performance requirements of MIL-C-28754, type III connectors unless otherwise specified herein.

19. Maintain 0.005 min electrical clearance between conductors and lands in Zone I. Conductor width in Zone I shall be maximized.

FIGURE 1. Dimensions and configuration - Continued.

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TOP VIEW

FIGURE 2. Flex Circuit contact assignment diagram.

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ELECTRICAL REQUIREMENTS:

Continuity testing: All Circuits shall have a resistance of no greater than 10 ohms when measured with a current of less than 3 amperes. This is a 100 percent test requirement.

Insulation resistance: Insulation resistance shall not be less than 99.9 megohms when tested in accordance with MIL-STD-1344. This is a 100 percent test requirement.

Dielectric withstanding voltage: The connector shall meet the requirements of MIL-STD-1344, Method 3001, for 500 Vac (rms) 60 Hz, or 700 Vdc. The point of measurement shall be between adjacent contacts.

Low level (dry circuits): Voltage drop shall not exceed 1.2 millivolts. Wire size is not applicable.

Contact resistance: Voltage drop shall not exceed 36 millivolts at 3 amps.

Contact retention: Per MIL-C-28754, the maximum displacement of the contact shall not exceed 0.005 with a pushout/pullout force of 4.0 plus or minus 0.1 lbs. There shall be no permanent displacement of the contact after removal of the force.

Current rating: 3 Amps max per contact and circuit trace (continuous) at 45°C max temperature rise.

Temperature rating: minus 55 to plus 125°C.

Voltage rating: 300 Volts, ac (rms), at sea level.

Ground contact: Install a contact grounding feature in contact location 25. The feature may incorporate an aluminum bushing and contact set or it may be of a compliant contact approach. All functional requirements for grounding bushings and contacts shall be per MIL-C-28754.

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MECHANICAL REQUIREMENTS:

Durability: The connector shall be subjected to 500 cycles of insertion and withdrawal at the maximum rate of 10 cycles per minute. After testing, the connector shall meet the qualification inspection requirements of table IV, Subgroup III of MIL-C-28754 and there shall be no exposure of nickel underplating on the connector contacts. The connector shall be attached to an appropriate frame/substrate assembly with all leads soldered into place. During the test, connectors shall be mounted so that the insertion and withdrawal forces are transmitted directly from the fin/header to the connector. Connectors shall be inserted into a backplane meeting the requirements of MIL-A-28870. The card cage test system shall provide a clearance of 0.007 plus or minus 0.001 inches (0.18 plus or minus 0.03 mm) between the surface of the guide rib and the interfacing rail surface of the card cage. Connectors shall be forced against the rail surface on each cycle after connectors are seated into the backplane. The clamping force shall be removed prior to extraction of the connectors. The clamping force shall be uniformly applied to the guide rib surface opposite the one in contact with the rail surface.

User Note: An encapsulant material around the leads is required to permit module assemblies to meet the durability requirements. The encapsulant must completely enclose the leads from the insulation to the PC Board.

Workmanship: Connector shall be inspected in accordance with the connector workmanship standard, drawing no. 53711-6133403.

Carrier strip: All solder leads shall be capable of being detached from the carrier strip in 5 plus or minus 2 bending cycles. The bending cycle is defined as the rotation of the carrier strip plus 30 to minus 90° from the nominal position.

Backplane contacts: "Low insertion force" contacts in accordance with MIL-C-28754/79, MIL-C-28754/80 and MIL-C-28754/81 are recommended for use with this connector.

Cleanliness: Connectors must meet the cleanliness requirement specified in MIL-P-28809.

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Insertion force: Force required to fully mate this connector with a corresponding backplane manufactured in accordance with MIL-C-28754/79, /80 or /81 shall be less than 18 pounds. Both backplane and connector shall be cleaned in freon TMS and allowed to air dry in a lab environment for a minimum of 30 minutes and a maximum of 4 hours prior to testing. Testing shall be performed in accordance with the assembly engagement and separation force procedure of MIL-C-28754. Backplanes used for this test shall be correlated with a reference standard maintained by NAVWPNSUPPCEN, Crane, Code 6042.

MATERIALS AND FINISHES:

Contacts: Half hard brass per QQ-B-613, Comp 260, or beryllium copper per QQ-C-533, 310 to 350 to the Knoop hardness scale.

Contact plating: Nickel underplate of 0.000050 inch minimum thickness per QQ-N-290, Class 1. Gold plating of 0.000050 inch minimum thickness per MIL-G-45204, Type II, Grade C, Class 1. Exposed base material is permitted in noncritical break-off plating carrier areas of the connector parts providing all performance requirements are maintained with no degradation of critical areas.

Insulators: Polyamide (nylon) per ASTM-D4066, Group I, Class 1, Grade 1, natural color. The contact ground feature can utilize an optional green nylon insulator.

Body shell: Aluminum per QQ-A-200/8, Type 6061-T6.

Body shell finish: Anodize per MIL-A-8625, Type III, Class 2, black. The inside surface of the 0.098 (2.49 mm) diameter holes shall not be anodized. Coating shall be 0.002 (0.05 mm) plus or minus 0.0005 (0.013 mm) thick on flat surfaces. All corners shall be covered. Corners shall have a slight radius to facilitate coverage.

Solder leads: Rolled half hard copper per IPC-CF-150E(CF-W6-2N or 3N), solder coated or tin-lead plated per MIL-P-81728 followed by fusing. Solder with Comp SN63 or SN60 of QQ-S-571, and the coating shall be 0.0003 to 0.001 thick. Leads shall meet the solderability

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requirements of MIL-STD-202, Method 208. The degree of exposed copper shall be in accordance with MIL-P-50884. The thickness of the copper leads in the flexcircuit before plating shall be in a range of 0.0025 to 0.0050 inches, except on the contact solder pad areas and unless otherwise specified on the drawing.

Flexcircuit: Per MIL-P-50884. Solder with Comp SN63 or SN60 of QQ-S-571 with quality requirements per MIL-P-50884. Apply conformal coating on the solder side of the flexcircuit. Insulation material shall be in accordance with IPC-FC-231/1 (polyimide) 0.001 nominal thickness.

Tab bushing: The sleeve shall have a minimum wall thickness of 0.010 inch and a minimum pushout force of 2.0 lbs. The sleeve end shall be flush within plus or minus 0.010 inch of the exterior surface of the connector tab and shall be a minimum of 0.07 inch long. A glass filled polytetrafluoroethylene material is suggested.

Solderability: In accordance with MIL-C-28754.

Soldering: In accordance with MIL-S-45743. The use of soldering methods other than hand soldering are acceptable provided the quality standards of MIL-S-45743 are applied.

Military part number: M28754/84- (Dash number from table I).

Qualification information: The connector shall meet the qualification test of MIL-C-28754, Table IV, Type IV, except for Subgroup IV, Fin or header torque, Fin cantilever load, and Guide rib stregth.

Number of units to be inspected: 12 of each dash number to be qualified shall be inspected.

Supersession data: Supersedes Naval Sea Systems Command Drawing Number 5516571, Code Identification Number 53711.

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TABLE I. Dash numbers.

DASH NUMBER	"X" DIMENSION	DIMENSION OF FINISHED TAB HOLE
01	0.000±0.010	WITH BUSHING INSTALLED Ø.070±.002 ID ⌖ Ø .010Ⓜ A BⓂ–CⓂ
02	0.025±0.010	
03	0.060±0.010	
04	0.000±0.010	WITHOUT BUSHING AS IN FIGURE Ø .100 $^{+.002}_{-.000}$ ⌖ Ø .007Ⓜ A BⓂ–CⓂ
05	0.025±0.010	
06	0.080±0.010	

Custodians:
NAVY - SH

User activities:
NAVY - AS

Agent:
NWSC, Crane

Preparing activity:
NAVY - SH

(Project: 5935-N336-06)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-28754/84	Connectors, Electrical, Modular, Type IV, Connector, 0.3 Pitch, 100 Contact, Offset

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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